

What is Python?

- Python is a Functional Programming Language
- Python is an Object-Oriented Programming Language
- Python is a High-Level Programming Language
- Python is a Interpreter Language
- Python is a scripting Language
- etc....



→ Python is a Programming Language which supports all above areas, so

Python is a General-Purpose Programming Language

→ What is Python?

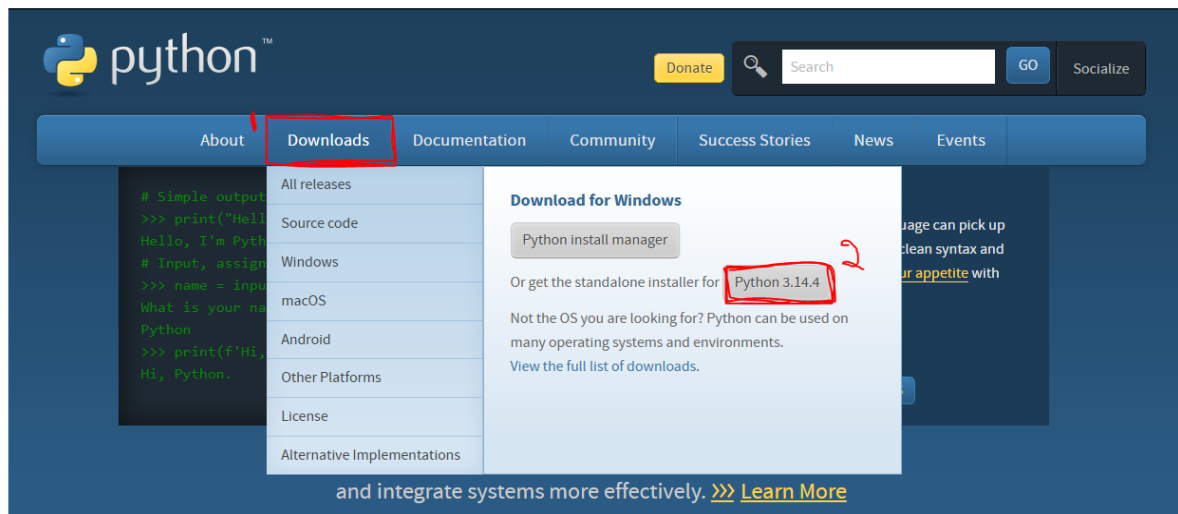
Python is a General-Purpose Programming Language developed by Guido van Rossum. The First version released in 1991.

Features of Python:

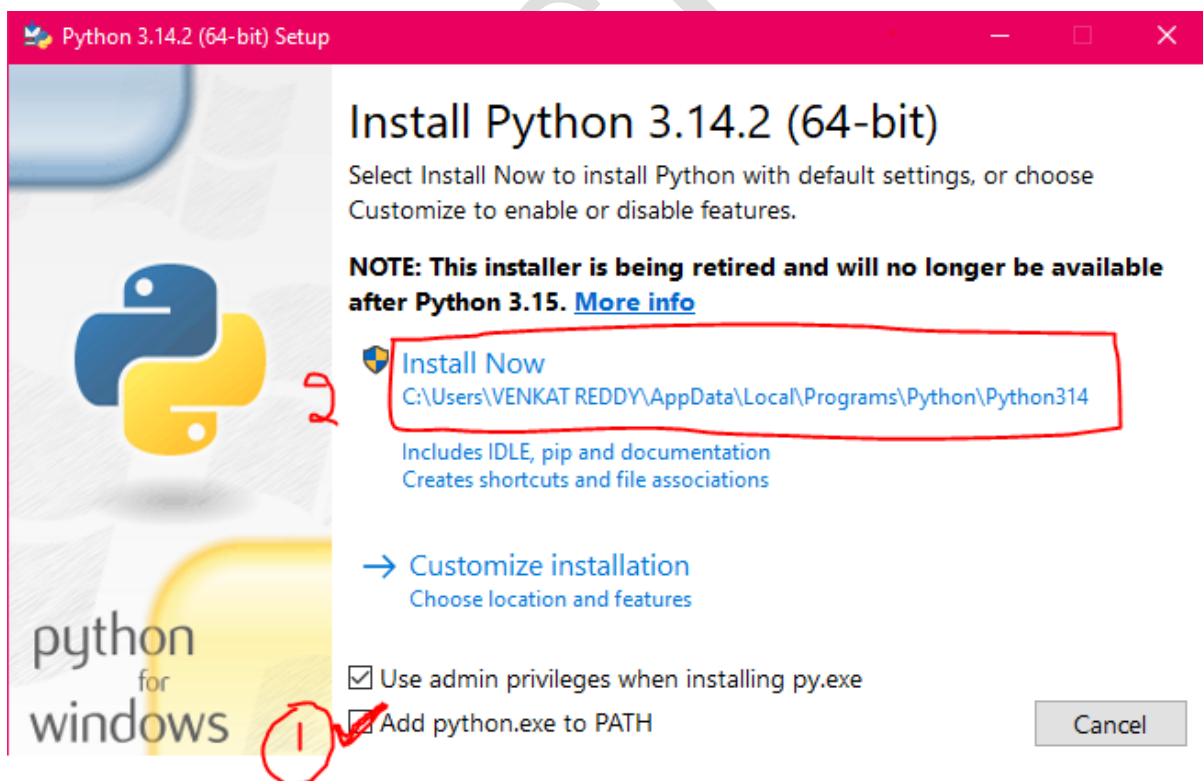
- 1. Open-Source and Free**
It is freely available on its official website
- 2. Easy to Code**
Compare to all other languages python code is very easy
- 3. Extensive Standard Library**
Python provides huge libraries so it is easy to handle code
- 4. Interpreted Language**
code is executed line by line instead of all at once
- 5. Dynamically Typed Language**
Python identifies data itself then store into variables

Steps to Install Python

Step1 : Download software from Python.org website



Step2: Once it's downloaded then double click on software

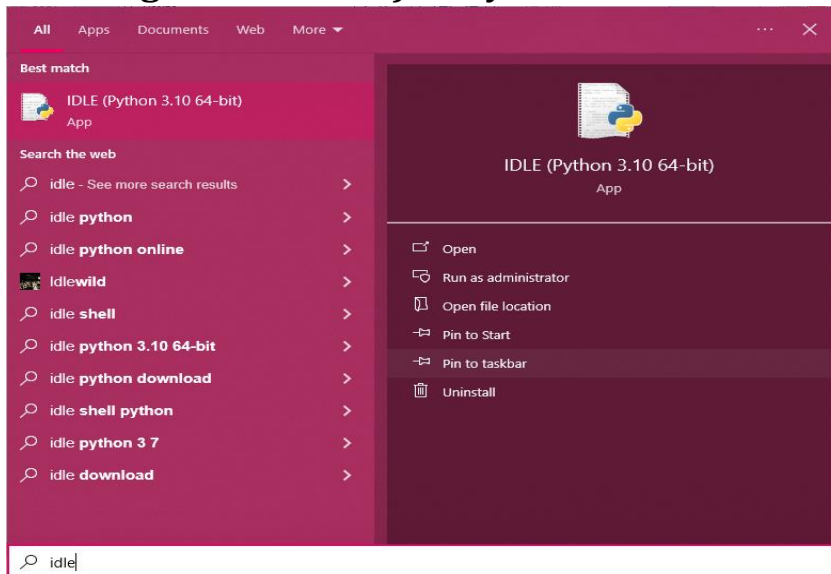


Give check mark for Add Python.exe to Path then select Install Now Option

Step3: Your Software is Installed in your system

Verify Python Software is Installed or not

➔ Select start button then type idle(Integrated Development Learning Environment) or Python



➔ If it appears that means python software is installed. (or)

➔ Open Command Prompt and type: Python -version

Working with Python

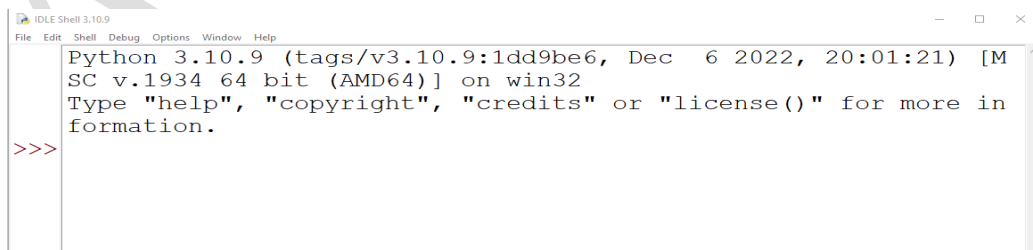
We can write scripts in 3 ways

1. By using Interactive Mode
2. By using Scripting Mode
3. By Using IDE's

1. **By using Interactive Mode:** For Every Line your will get output

Start ➔ Type IDLE ➔ Press Enter

Now you will get a window which is called as IDLE-Interactive Mode



2. **By using Scripting Mode:**

We can write group of lines then we can save and execute script

Start ➔ Type IDLE ➔ Press Enter

By default IDLE-Interactive mode opens, now

Select File ➔ Choose new file option ➔ Write Script ➔ Save it

➔ Choose Run option

3. By using IDE's(Integrated Development Environment)

We have different types of IDE's Available, They are

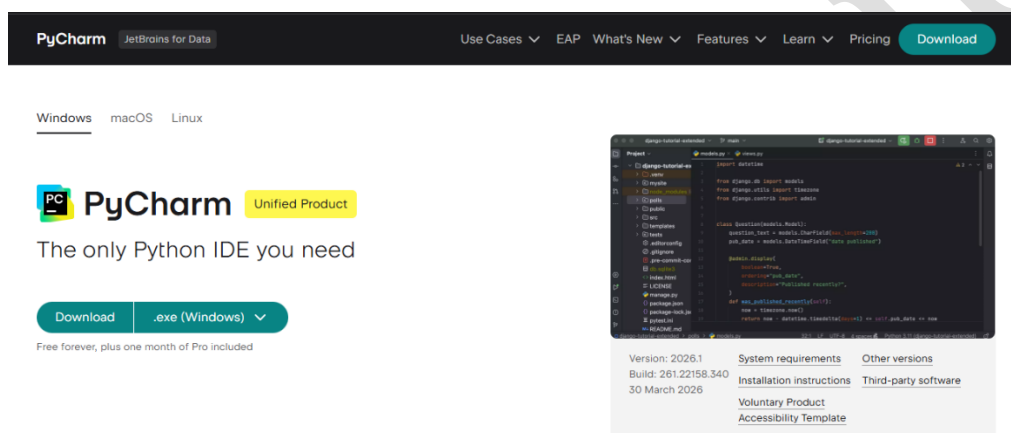
1. Pycharm
2. VsCode
3. Jupyter
4. Google CoLab
5.

Steps to Install Pycharm IDE's

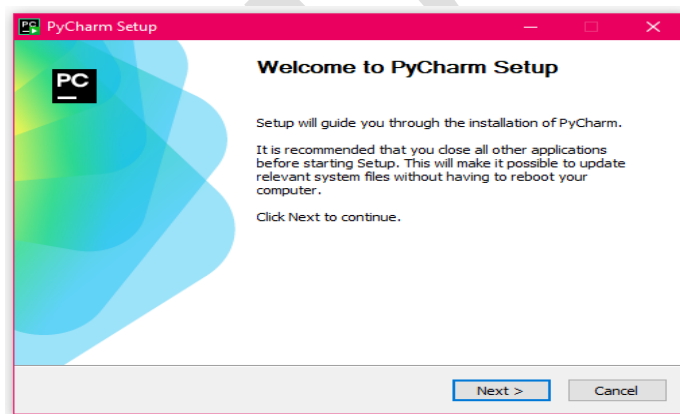
Step 1: Download PyCharm

Open browser and go to

👉 <https://www.jetbrains.com/pycharm/download/>



Step2: Go to location where you downloaded then double click on software



Select Next option

Choose installation location → Click Next

In Installation Options, select:

- ☒ Add "Open Folder as Project"
- ☒ Create Desktop Shortcut
- ☒ Add launchers dir. to PATH
- (Optional but useful)
- Click **Install**
- After installation → Click **Finish**

Step 3: First Launch Setup

1. Open PyCharm
2. Choose:
 - Do not import settings (for first time)
3. Select UI theme (Light/Dark)
4. Click Skip remaining and set defaults

Step4: Configure Python in PyCharm

1. Open PyCharm
2. Click New Project
3. You will see "Python Interpreter" section
Choose Interpreter:
 - Select **New Virtual Environment**
 - OR choose **Existing Interpreter**
4. Click **Create**

Step 6: Run First Python Program

1. Right-click project → New → Python File
2. Name it test.py
3. Write: `print("Hello World")`
4. Right-click → Run

Realtime Applications by using Python:

- ➔ Instagram and Spotify developed by using Python
- ➔ Netflix uses Python for Analyzing user Data
- ➔ Google uses python for AI models
- ➔ Daily Report automated in companies like sending emails
- ➔ Moving data from Database-->cleaning-->loading into BI tools
- ➔ Etc.